MILITARY SPECIFICATION

IGNITER, MARK 188 MOD 0

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE.

1.1 Scope. - This specification covers the requirements for the igniter, Mk 188 Mod 0. This is the only classification covered by this specification.

APPLICABLE DOCUMENTS

The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-124	Containers (Cans, Parts and Drums) Metal

BUREAU OF NAVAL WEAPONS

OS 10377	Sealant for Solid propellant Rocket Motors
OS 10379	Ignition Material for Solid Propellant Rocket Motors
OS 11334	Ignition Element, Mk 8 Mod 0

FSC 1340

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables

DRAWINGS

BUREAU OF NAVAL WEAPONS

2185553	Palletized Shipping Container
2185554	Unit Pack, Inside Shipping Container
2185555	Can, Inside Shipping Container
2185556	Box, Outside Shipping Container
LO 537849	Igniter, Rocket Motor Mark 188 Mod 0 Assembly, List of Drawings, Assemblies, Parts, and Specifications

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(When requesting any of the applicable documents, refer to both title and number. , Requests should be made via the cognizant Government Inspector. Copies of specifications and applicable documents required by contractors are available upon application to the Commanding Officer, Naval Supply Depot (Code CDS), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, Copies of OS specifications and Ordnance drawings should be requested from the Commanding Officer, Naval Ordnance Depot, Louisville 14, Kentucky, Attention: CTDO.)

2.2 Other Publications. - The following document forms a part of this specification, unless otherwise indicated, the Issue in effect on date of invitation for bids shall apply to the extent specified herein.

INTERSTATE COMMERCE COMMISSION

49 CFR 71-78	Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition - Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D .C. Orders for the above publication should cite “49 CFR 71-78 (Rev. 1950) ”.)

3: REQUIREMENTS

3.1 Preproduction Lot. - Unless otherwise specified by the procuring activity, the contractor s a , before beginning large scale production under any contract or order, submit for testing in accordance with 4.4, a preproduction lot consisting of 31 igniters manufactured by the procedures proposed for use in the execution of the contract or order. Should the preproduction lot fail to comply with the requirements of this specification, the procuring activity may require the submission of addi- - tional preproduction lots.

3.2 Materials. - Materials shall comply with the applicable drawings and specifications as shown on LD 537849. Substitution of materials may not be made without prior consent of the procuring activity. The use of substitute materiels shall not, however, constitute a waiver of any requirement or inspection provision of this specification.

3.2.1 Ignition Element. - The ignition element, Mk 8 Mod O, shall comply with the requirements 1334,

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3.2.2 Pelletized Pyrotechnic composition. - The pelletized pyrotechnic composition shall comply with the requirements of OS 10379.

3.2.3 Cotton. - Cotton packing shall be long staple. Moisture content shall not exceed 2.0 percent. The cotton shall I be free from foreign material.

3.2.4 Basket Coating. - The wire mesh basket of the body and basket assembly shall be completely coated with a coating solution, except for the opening in the bottom of the basket which is normally used for loading. This coating shall I be adequate to provide a moisture - proof seal after the closure plug is cemen- - ted in place. The basket coating solution shall comply with the requirements of OS 10377.

3.3 Electrical Resistance. - The resistance of the igniter circuit measured between the igniter grounding post and the igniter cable assembly shall I be 0.07 to 0.15 ohms.

3.4 Performance. - When the igniter is fired in a closed chamber in accordance with 4.6.2, the results shall comply with the following requirements:

3.4.1 The time to first indication of pressure shall I be no greater than 0.015 seconds.

3.4.2 The time required to attain a gage pressure of 200 pounds per square inch shall be 0.045* 0.015 seconds. When tested from -85°F to +175°F.

3.4.3 The maximum gage pressure obtained in the chamber shall I be between 400 and 650 pounds per square inch.

3.5 Workmanship. - The workmanship shall be that required by the best industrial practices governing the quantity production of analogous parts, maintaining the dimensions, finishes, tolerances, and quality specified herein and on applicable drawings, specifications and inspection manuals

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of al I inspection requirements as specified herein. Except as otherwise specified, the supplier may

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utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. - A lot shall consist of all units manufactured by a uniform process in which a homogeneous product would be expected. The maximum lot size may be specified at the option the procuring activity.

4*3 Classification of Tests. - The inspection and testing of units shall be classified as follows:

4.3.1 Preproduction Test. - The preproduction test is that group of tests conducted prior to start of production to demonstrate the suitability of the unit for production and service use.

4.3.2 Acceptance Test, - The acceptance test is that group of tests conducted on units submitted for Acceptance under contract to demonstrate suitable end item quality control, correct assembly, and performance .

4.4 Preproduction Tests. - Unless otherwise specified in the contra or order a preproduction lot of 31 igniters shall be subjected to tests in the quantities indicated and in accordance with the procedures stated herein.

4.4.1 Circuit Resistance. - The electrical resistance of the igniter circuit shall be determined on each unit and shall comply with 3.3.

4.4.2 Non-functioning Test. - Each igniter shall have a current of 2.00 (+0.01, -0.00) amperers applied for 60 seconds minimum. No igniter shall function with this current applied.

4.4.3 Statistical Tests. - Fifteen igniters shall be selected at random from the preproduction lot and divided into three groups of five igniters each, designated Groups 1, II, and III. The igniters shall be fired in accordance with 4.6.2 at the following temperatures:

Group I	-	-85°F
Group II	-	60°F
Group III	-	175°F

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The igniters shall meet the performance characteristics specified in 3.4,

4.4.4 Sequential Tests. - Eight igniters shall be selected at random from the preproduction lot and divided into two groups of four igniters each, designated Groups IV and V. The igniters shall be subjected to tests as Indicated in the follow- - ing paragraphs.

4.4.4.1 Group IV.

4.4.4.1.1 Vibration Tests. - Each igniter in Group IV shall be conditioned at -85°F and vinrated at -85°F with o nominal frequency of 30 cycles per second at an applied double amplitude of 0.060 ± 0.002 inch peak to peek for eight hours at each of the three mutually perpendicular axis. Each igniter shall I then be conditioned to 175°F and vibrated at 175°F with a nominal frequency of 30 cycles per second at an applied double amplitude of 0.060 ± 0.002 inch peak to peak for eight hours at each of the three mutually perpendicular axis.

4.4.4.1.2 Temperature Cycling Tests. - Following vibration each igniter shall be temperature cycled from -85°F to 175°F for three complete cycles. Each unit shall be conditioned at each successive temperature a minimum of eight hours, with a maximum time of 10 minutes between temperature changes.

4.4.4.1.3 Performance Test. - After completion of the last cycle each igniter shall be fired at -85°F in accordance with 4.6.2. The performance of each unit shall comply with the requirements of 3.4.

4.4.4.2 Group V.

4.4.4.2.1 Vibration Tests. - Each igniter in Group V shall be conditioned at 175°F and vibrated at 175°F with a nominal frequency of 30 cycles per second at an applied double amplitude of 0.060 * 0.002 inch peak to peak for eight hours at each of the three mutually perpendicular axis. Each igniter will II then be conditioned to -85°F and vibrated at -85°F with a nominal frequency of 30 cycles per second at an applied doubt e amplitude of 0.060 ± 0.002 inch peak to peak for eight hours at each of the three mutually perpendicular axis.

4.4.4.2.2 Temperature Cycling Tests. - Following vibration each igniter shall be temperature cycled from 175°F to -85°F for three complete cycles. Each unit shall I be conditioned at each successive temperature a minimum of eight hours, with a maximum time of ten (10) minutes between temperature changes.

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4.4.4.2.3 Performance Test. - After completion of the last cycle each igniter shall be fired at 175°F in accordance with 4.6.2. The performance of each unit shall comply with the requirements of 3.4.

4.4.5 Accelerated Aging Tests. - Eight igniters shall be randomly selected from the preproduction lot and divided into two groups of four igniters each, designated Groups VI and VII. Each igniter in Group VI shall be stored at 175°F for thirty days, then fired at 175°F in accordance with 4.6.2. Each igniter in Group VII shall be stored at [illegible]°F for 30 days, then fired at -85°F in accordance with 4.6.2. The performance of each igniter from Groups VI and VII shall [illegible] with the requirements of 3.4.

4.4 Preproduction Test Report. - Following completion of all preproduction tests a report shall be submitted which will constitute a record of all information pertaining to the igniter tests. This report will normally be used as a basis for approval of the preproduction test, and shall contain, but not necessarily be limited to, the following:

(a) Description of tests
(b) Chronological history of all events connected with testing
(c) Analysis of results
(d) Conclusions and recommendations
(e) All test data

4.5 Acceptance Tests.

4.5.1 Nondestructive Testing.

4.5.1.1 Examination. - Each igniter delivered or tested shall be examined to assure quality of workmanship, use of proper materials and assembly, and absence of visual defects or other imperfections that would affect the life or function of the unit.

4.5.1.2 Electrical Resistance. - The electrical resistance of each igniter delivered or tested shall be determined and shall comply with the requirements of 3.3.

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4.5.1.3 Dimensional Inspection. - A sample shall be selected from each lot in accordance with the provisions of MIL -STD-I05A, inspection level II, normal inspection, single sampling. The sample shall be inspected for compliance with applicable dimensional requirements of drawings listed in LD 537849 to an Acceptable Quality Level (AQL) of 1.5.

4.5.2 Performance Tests. - A sample shall I be selected from each lot, test at -85°F and +175°F, in accordance with the provisions of Ml L-STD-414, inspection level IV, normal inspection with variability unknown -standard deviation method. Each unit shall be tested in accordance with 4.6.2 and shall comply with the requirements of 3.4. Failure of the sample to comply with the acceptability limits for on AQL of 0.15 shall constitute rejection of the lot.

4.6 Test Methods.

4.6.1 Electrical Resistance Test. - The resistance and continuity of the completed igniter shall I be measured between the igniter grounding post and the igniter cable assembly. A resistance testing circuit which subjects the igniter to less than 10 milliamperes shall I be used. Resistance measurements shall be made with a suitable meter having an accuracy of ° 0.005 ohms.

4.6.2 Performance Test.

4.6.2.1 The igniter shall be installed in an igniter test chamber fabricated in accordance with Figure 1 and having a free volume of 610 ± 6 cubic inches. The igniter shall be fired electrically with not less than 15 nor more than 20 amperes of current flowing l

4.6.2.2 A record of chamber pressure versus time shall be obtained for each test in the Igniter test chamber. This record shall have a minimum time resolution of 0.001 second and minimum pressure resolution of 10 psi and shall consist of a continuous recording from the instant that current is applied to the igniter until maximum pressure is attained.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 In addition to the following Level A packaging shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

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5.101.1 Unit Packaging. - The igniter shall be packaged in a metal container fabricated in accordance with BuOrd Drawing 2185555. THe container shall comply with the requirements of MI L-C- 124B, Type Ill.

5.2 Packing.

5.2.1 Level A. - In addition to the following Level A packing shall be in accordance with the Code of Federal Regulations 49 CFR 71-78. The package material and method of packaging shall I comply with BuOrd Drawing- 2185554.

5.2.1.1 Exterior Containers, - The unit containers shall be in accordance with BuOrd Drawing 2185556. These exterior containers shall then be pal letized in accordance with BuOrd Drawing 2185553, except that shipments of small quantities which do not warrant pallet size loads may be shipped in the exterior container only. Adequate bracing shaould be assured to prevent movement or damage in transit.

5.3 Marking.

5.3.1 Special Markings. - Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.3.2 Normal Markings. - In addition to the markings required by contract or order, unit packages and shipping containers shall I be marked in accordance ‘with the requirements of MIL -STD-126.

5.4 Storage. - Igniters shall be capable of storage in any position for a period of three-n a storage temperature range of -40°*F to 140°F without deterioration.

6. NOTES

6.1 Intended Use. - Igniters covered by this specification are intended to be used to ignite the Rocket Motor Mk 25 Mods 0, 1 and 2.

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6.2! Ordering Data. - Procurement documents should specify the fol - lowing:

(a) Full name of item, including Mk and Mod numbers
(b) Title, number, and date of this specification
(c) Preproduction lot size, when required
(d) Maximum lot size, when required .
(e) Drawing numbers and other applicable documents

6.3 Safety Precautions. -The loading, assembly and handling of the igniter and relate explosive components thereof, covered by this specification involve hazardous operations and therefore require the observance of suitable explosives safety precautions. Use of this specification should not be construed to relieve the contractor or the manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions which a contractor or manufacturer of the igniter should observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons and other Government plants these provisions are mandatory. Such other warnings and precautions, which effect the operatin effectiveness or safety during use of the igniter, are included in the detail provisions of this specification.

6.3.1 All loading operations should be conducted in a neat and orderly manner.

6.3.2 Personnel handling the loaded igniter and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion or deterioration from perspiration or other contaminating deposits.

6.3.3 The exposure of explosive materials and related park should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.3.4 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD). While in process, assemblies and subassemblies should be stored in safety lockers or chests when in loading rooms. All storage should be in adequate ready for service magazines, located in accordance with intraplant distances, when outside of loading rooms.

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6.3.5 Safe equipment methods should be utilized for transporting and handling explosives and loaded parts.

6.3.6 Proper care should be exercised at all times to protect personnel from accidents, fires, or explosions and to limit damage to equipment and loading areas. in this connection, the following precautionary measures should be observed:

6.3.6.1 Keep only minimum quantities of explosives and completed or partial loaded parts present at each stage of operation.

6.3.6.2 Keep explosives and explosive parts in approved, covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive in order to ground electrostatic charges,

6.3.6.3 Protect all operations from electrostatic charges by effectively grounding all machinery, equipment and fixtures and, where necessary, provice , suitable grounded conductive coverings on f floors, work benches, tables and workers' shoes l

6.3.6.4 Enforce, where necessary, the wearing of suitable footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning, and similar hazards.

6.3.6.5 Allow no fires or exposed sparking of electrical equipment and little or no inflammable material to be present. in boding, handling and storage spaces Enforce proper “Match” and "No Smoking” rules where necessary. “

6.3.6.6 Enforce good housekeeping and policing methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosive material, and assure its removal from floors, walls, ceilings, ledges, tables, benches and work equipment. Clean up spilled material immediately.

6.3.6.7 Employ properly proportioned and properly located protective. barricades, screens, or shields at all required points.

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6.3.6.8 Motors switches and relays for equipment should be explosion proof and adequately grounded.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor
* any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manu - facture, use or sell any patented invention that may l be related thereto.

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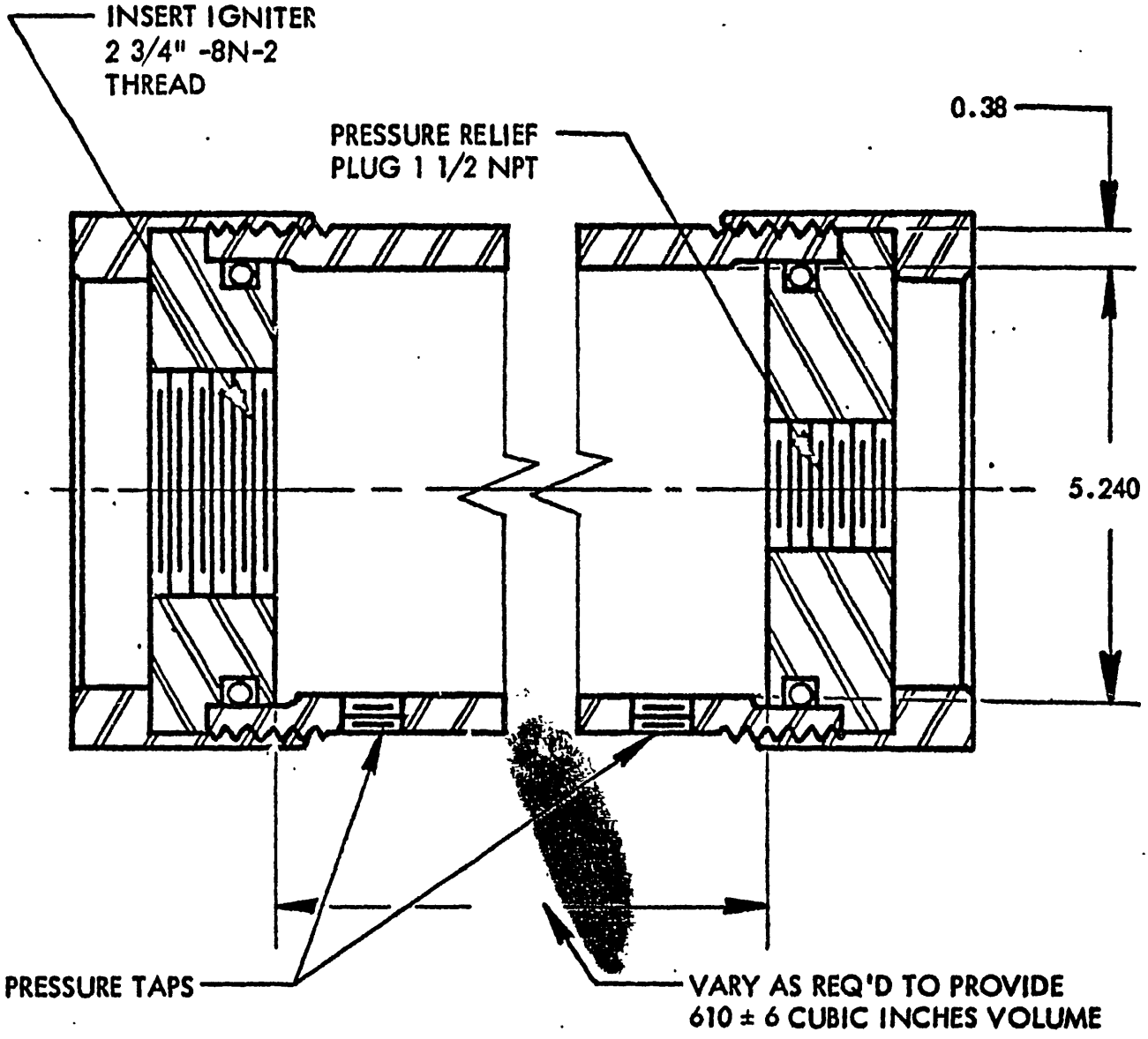


FIGURE 1

PERFORMANCE TEST CHAMBER

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